

Birdly E-Commerce Application

Contents

# Project Overview	2
## Purpose	2
# Technology Stack	2
# Product Features	2
## Product Listing	2
## Capabilities	2
## Shopping Cart and Data Management	2
### Capabilities	2
## Backend Server	3
### Capabilities	3
## Challenges and Solutions	3
### Challenge 1: CODIO Deployment	3
### Challenge 2: PowerShell Execution Policy	3
### Challenge 3: Mobile Testing	4
### Challenge 4: Invalid Date Display	4
## Testing and Validation	4
### Functional Testing	4
### Browser Testing	4
### User Acceptance Testing	5
## Future Improvements	5
### Planned Enhancements	5
## Repository and Live Links	5
### Final Project Repository	5
### Frontend Repositories	5
### Backend Repositories	5
### Frontend Live Links	6
### Backend Project Links	6
### Local Testing	6

Birdly E-Commerce Application

Project Overview

Purpose

The Birdly E-Commerce Application provides a centralized platform for bird-watching enthusiasts and collectors to discover and purchase bird-themed merchandise.

Technology Stack

Layer	Technologies
Frontend	HTML, CSS, JavaScript
Pages	productList.html, orders.html, checkout.html
Backend	Node.js, Express.js, server.js, jsfile.js
Database	SQLite
Hosting	Local Development Environment

Product Features

Product Listing

The main page displays available bird-themed products.

Capabilities

- View products
- Add items to cart

Shopping Cart and Data Management

Supports CRUD operations and transactional processing.

Capabilities

- Create cart records

Birdly E-Commerce Application

- Read cart contents
- Update quantities
- Complete checkout workflow
- Create orders

Backend Server

Provides server-side functionality using Node.js and Express.

Capabilities

- Serve frontend pages
- Handle API requests
- Connect to SQLite
- Support localhost testing

Challenges and Solutions

Challenge 1: CODIO Deployment

Problem

The application worked on localhost but failed to add items to the cart when deployed to CODIO.

Solution

Worked with the instructor to debug the deployment and continued testing locally.

Outcome

Successfully deployed and tested on localhost.

Challenge 2: PowerShell Execution Policy

Problem

Node.js setup required administrator permissions.

Solution

Used the CurrentUser execution policy scope instead of LocalMachine.

Outcome

Successfully configured the development environment without administrator privileges.

Birdly E-Commerce Application

Challenge 3: Mobile Testing

Problem

Buttons did not respond on iPhone Safari.

Solution

Changed localhost references to the laptop's local IPv4 address.

Outcome

The application functioned correctly on iPhone devices.

Challenge 4: Invalid Date Display

Problem

Order dates displayed as "Invalid Date."

Solution

Updated date formatting:

```
new Date(order.created_at).toLocaleString()
```

Outcome

Dates displayed correctly.

Testing and Validation

Functional Testing

Test Case	Result
View Product List	Passed
Add Item to Cart	Passed
Update Cart Quantity	Passed
Checkout Process	Passed
Order Creation	Passed
SQLite Database Operations Passed	

Browser Testing

- Microsoft Edge

Birdly E-Commerce Application

- Google Chrome
- Mozilla Firefox
- iPhone Safari

User Acceptance Testing

The application was tested by family members using iPhone Safari and all primary functionality worked successfully.

Future Improvements

Planned Enhancements

- Improve mobile responsiveness
- Add product search and filtering
- Optimize performance
- Add user account management
- Implement analytics dashboards
- Track product popularity and conversions

Repository and Live Links

Final Project Repository

GitHub Repository:

https://github.com/labricot-lab/446final_20260611

Live Site:

https://labricot-lab.github.io/446final_20260611/

Frontend Repositories

https://github.com/labricot-lab/m5_lab5_20260310.git

https://github.com/labricot-lab/446_m5_coursepro_20260312.git

Backend Repositories

https://github.com/labricot-lab/446_pro_08_20260421.git

Birdly E-Commerce Application

https://github.com/labricot-lab/446_pro_07_20260419.git

https://github.com/labricot-lab/446_pro8_updated_20260506.git

Frontend Live Links

https://labricot-lab.github.io/m5_lab5_20260310/

https://labricot-lab.github.io/446_m5_coursepro_20260312/

Backend Project Links

https://labricot-lab.github.io/446_pro_08_20260421/

https://labricot-lab.github.io/446_pro_07_20260419/

https://labricot-lab.github.io/446_pro8_updated_20260506/

Local Testing

<http://localhost:3000>

<http://192.168.0.53:3000>